

## LAB 2: AGILE SCRUM PROJECT MANAGEMENT USING JIRA

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Project title : Pharmacy Expiry Management System

### 1. Aim

To implement and simulate an Agile Scrum project using Jira by creating Epics and User Stories, prioritizing the backlog, assigning Story Points, creating and executing two Sprints, tracking progress, and analysing the Sprint Burndown Chart.

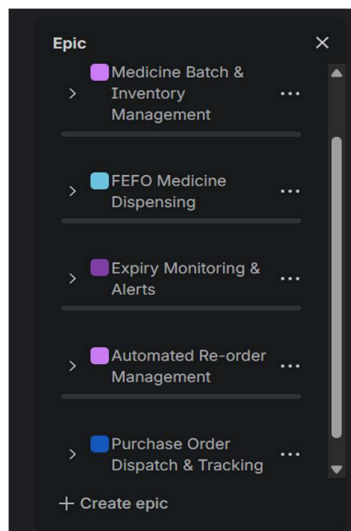
### 2. Project Description

The project is a Pharmacy Expiry Management System designed to help pharmacies manage medicine batches, dispense medicines using the FEFO principle, monitor medicine expiry, automatically reorder medicines, and track purchase orders.

### 3. Epics

| Epic                                  | Purpose                                                               |
|---------------------------------------|-----------------------------------------------------------------------|
| Medicine Batch & Inventory Management | Manage medicine batches, quantities and inventory information         |
| FEFO Medicine Dispensing              | Dispense medicines according to First-Expire-First-Out                |
| Expiry Monitoring & Alerts            | Identify near-expiry medicines and generate expiry alerts             |
| Automated Re-order Management         | Automatically generate purchase orders when stock needs replenishment |
| Purchase Order Dispatch & Tracking    | Dispatch purchase orders and track their status                       |

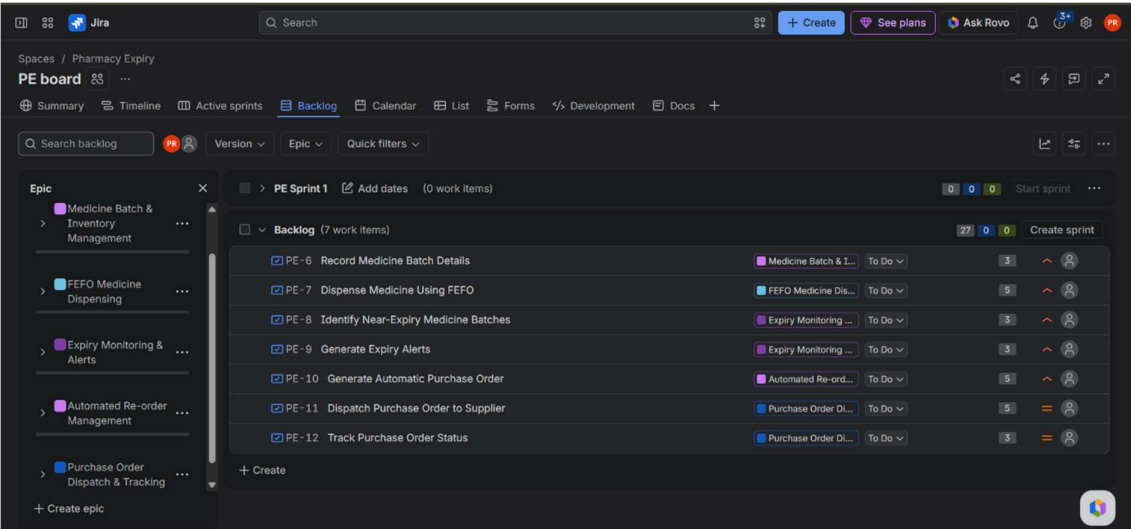
Figure 1: Epics created for the Pharmacy Expiry Management System in Jira



4. User Stories

| User Story                            | Epic                                  | Priority | Points |
|---------------------------------------|---------------------------------------|----------|--------|
| Record Medicine Batch Details         | Medicine Batch & Inventory Management | High     | 3      |
| Dispense Medicine Using FEFO          | FEFO Medicine Dispensing              | High     | 5      |
| Identify Near-Expiry Medicine Batches | Expiry Monitoring & Alerts            | High     | 3      |
| Generate Expiry Alerts                | Expiry Monitoring & Alerts            | High     | 3      |
| Generate Automatic Purchase Order     | Automated Re-order Management         | High     | 5      |
| Dispatch Purchase Order to Supplier   | Purchase Order Dispatch & Tracking    | Medium   | 5      |
| Track Purchase Order Status           | Purchase Order Dispatch & Tracking    | Medium   | 3      |

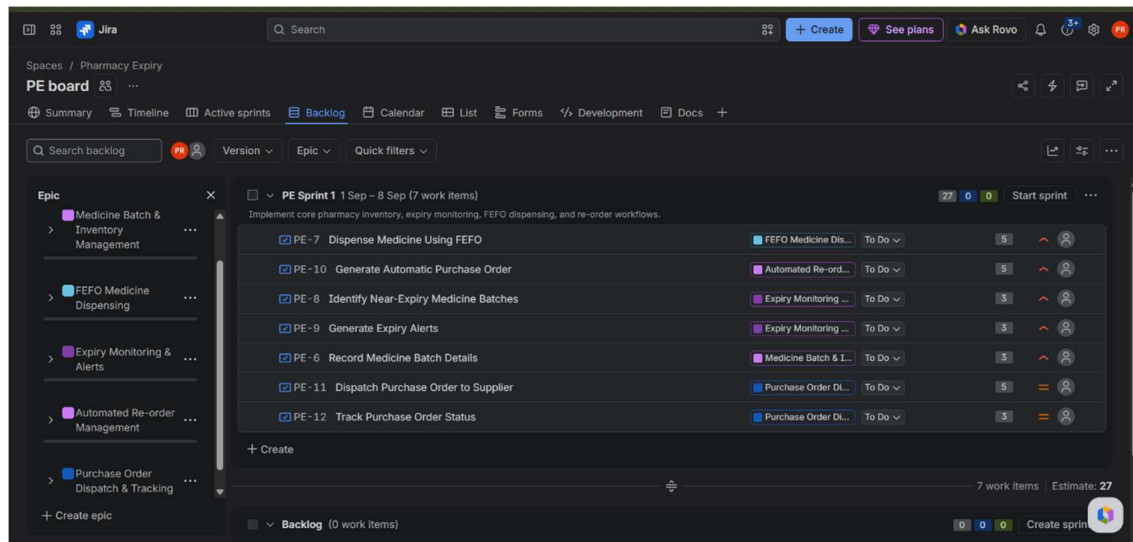
Figure 2: User Stories populated under their respective Epics in the Jira backlog



5. Prioritization of Backlog

The User Stories were prioritized according to their importance and user impact.

Figure 3: Prioritized Product Backlog

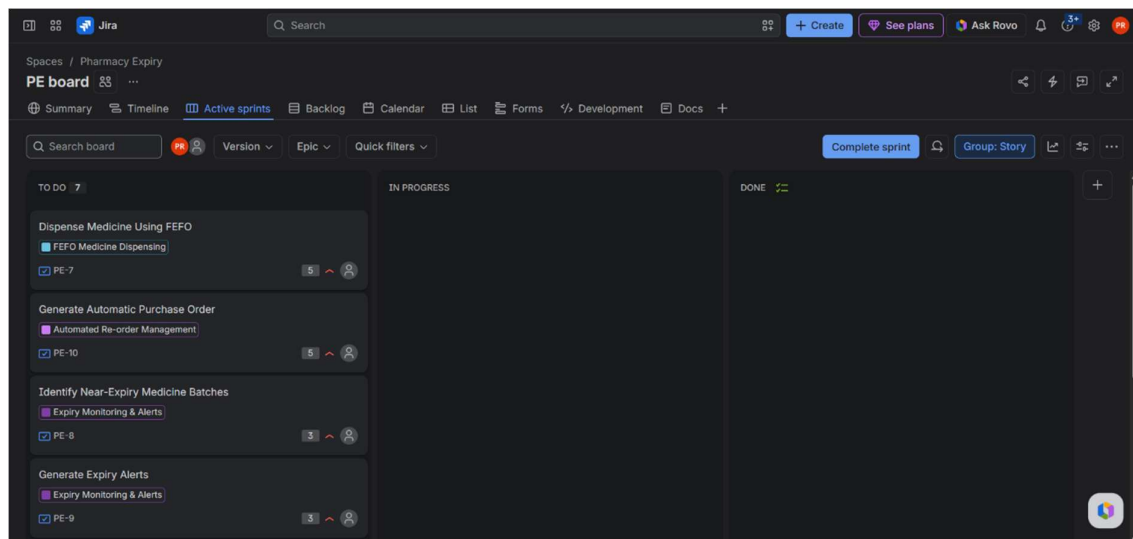


## 6. Sprint 1 Planning

A one-week Sprint 1 was created. The selected User Stories were moved from the backlog into PE Sprint 1.

Sprint Goal - Implement core pharmacy inventory, expiry monitoring, FEFO dispensing, and re-order workflows.

Figure 4: Sprint 1 planned with selected User Stories

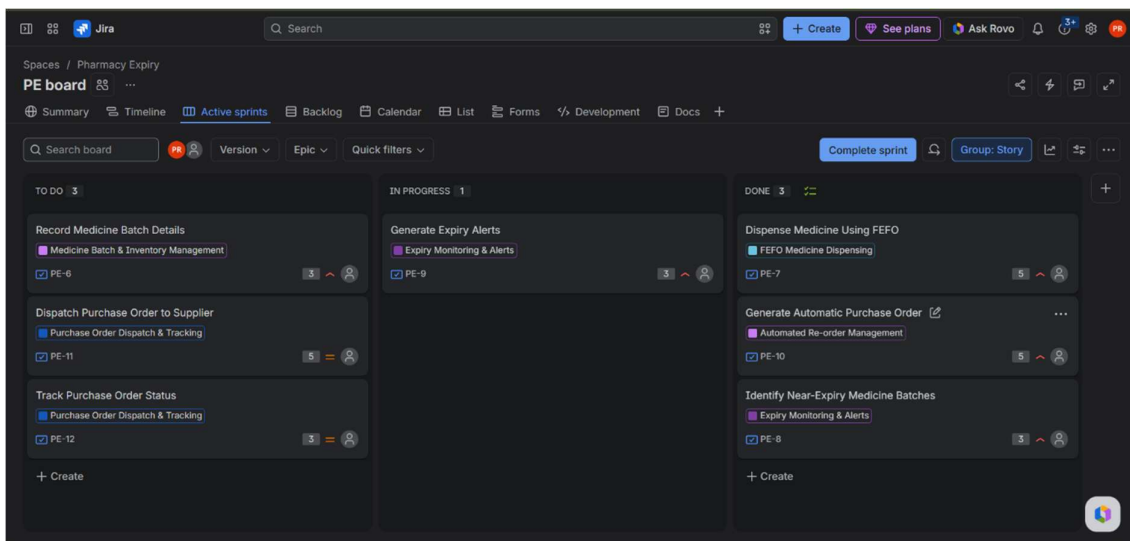
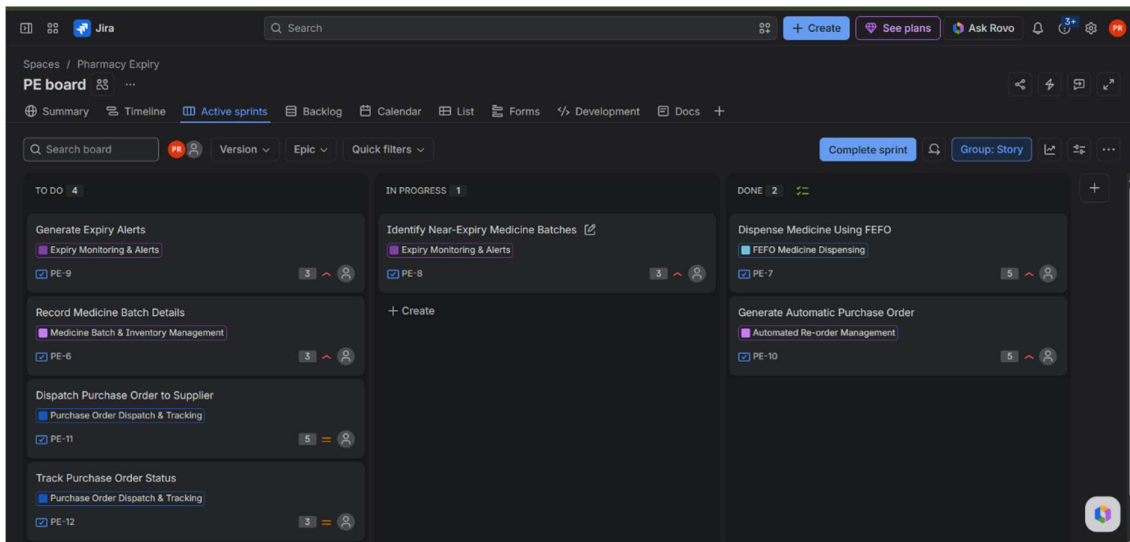
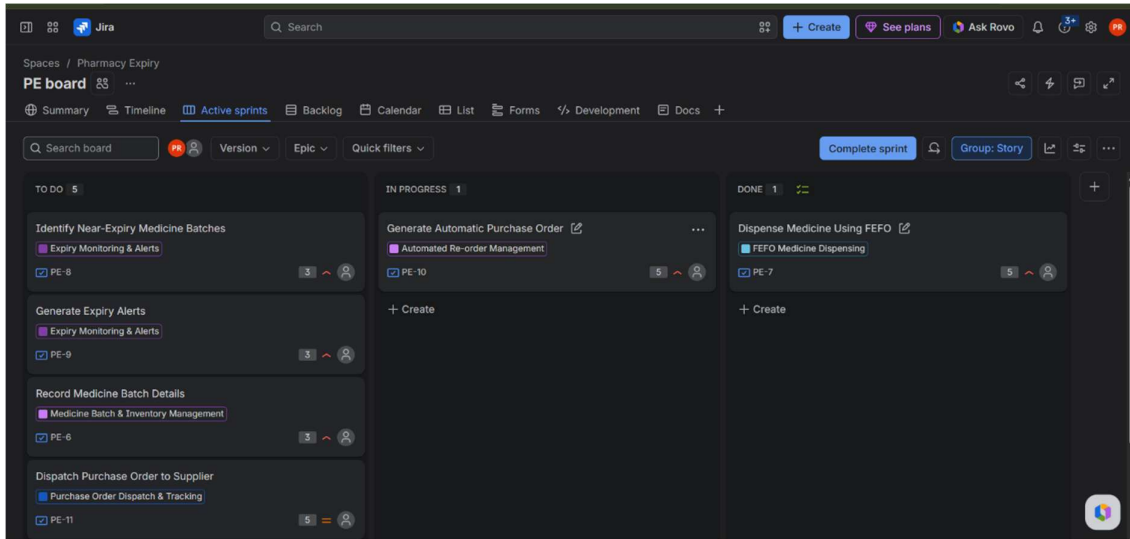


## 7. Sprint 1 Execution

Sprint 1 was started and the User Stories were moved through the Scrum workflow:

To Do → In Progress → Done

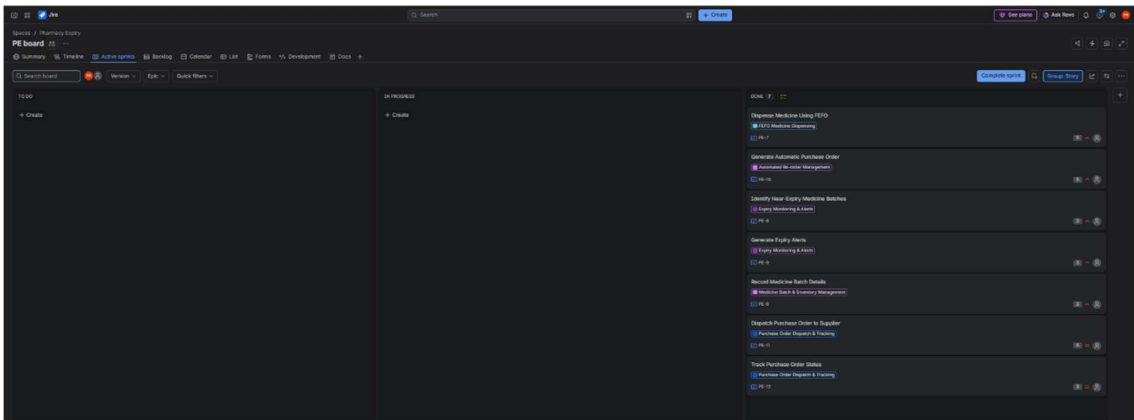
Figure 5: Sprint 1 progress showing User Stories in To Do, In Progress and Done states



## 8. Sprint 1 Completion

All seven User Stories in Sprint 1 were moved to Done.

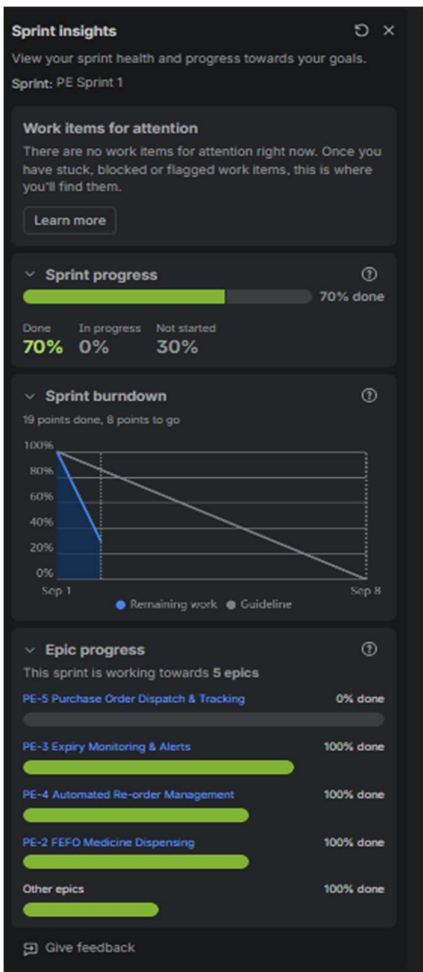
Figure 8: Completion of all User Stories in Sprint 1



9. Sprint 1 Insights

Jira's Sprint Insights showed the progress of the sprint.

Figure 9: Sprint Insights showing Sprint 1 progress, burndown and Epic progress



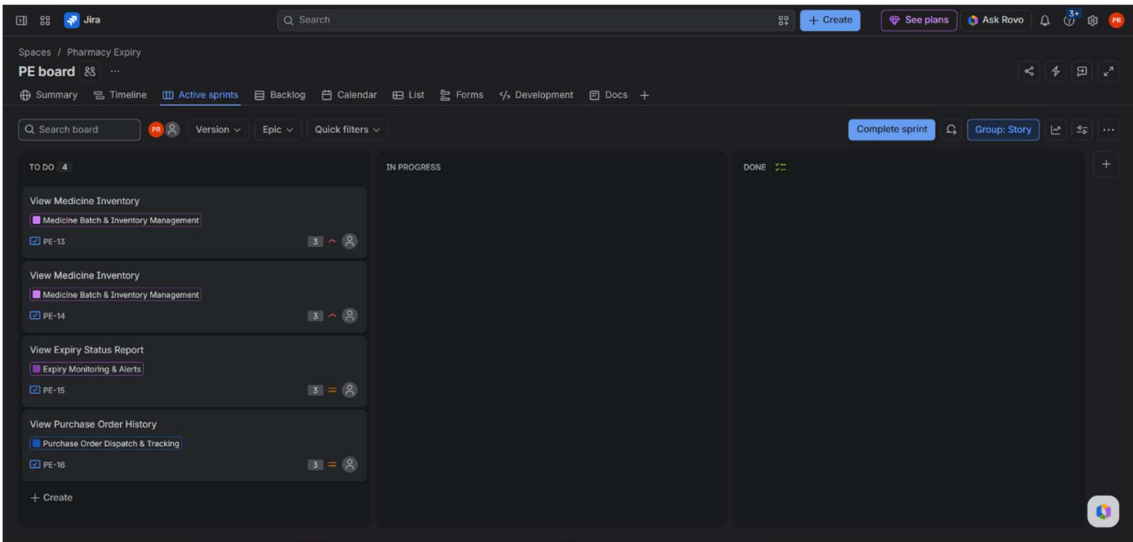
## 10. Sprint 2 Planning

Sprint 2 Stories :

| User Story                  | Epic                                  | Priority | Points |
|-----------------------------|---------------------------------------|----------|--------|
| View Medicine Inventory     | Medicine Batch & Inventory Management | High     | 3      |
| View Expiry Status Report   | Expiry Monitoring & Alerts            | Medium   | 3      |
| View Purchase Order History | Purchase Order Dispatch & Tracking    | Medium   | 3      |

Sprint 2 Goal - Improve pharmacy inventory visibility and procurement tracking by providing inventory status, expiry reports, and purchase order history, enabling pharmacy staff to make informed decisions and manage medicines efficiently.

Figure 10 : Sprint 2 planned with additional User Stories

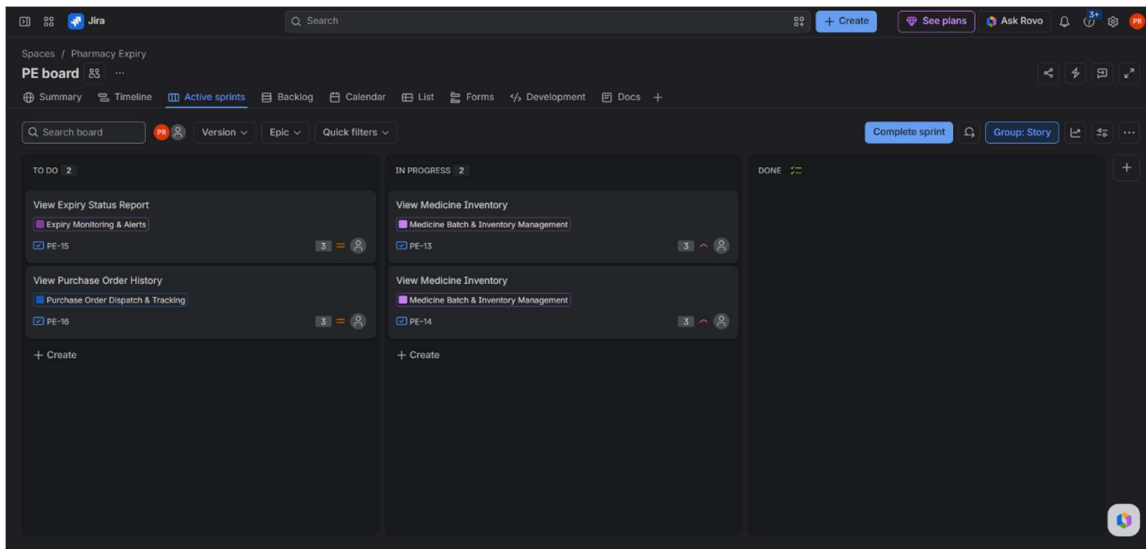


## 11. Sprint 2 Execution

The Sprint 2 User Stories were simulated through the Scrum workflow:

To Do → In Progress → Done

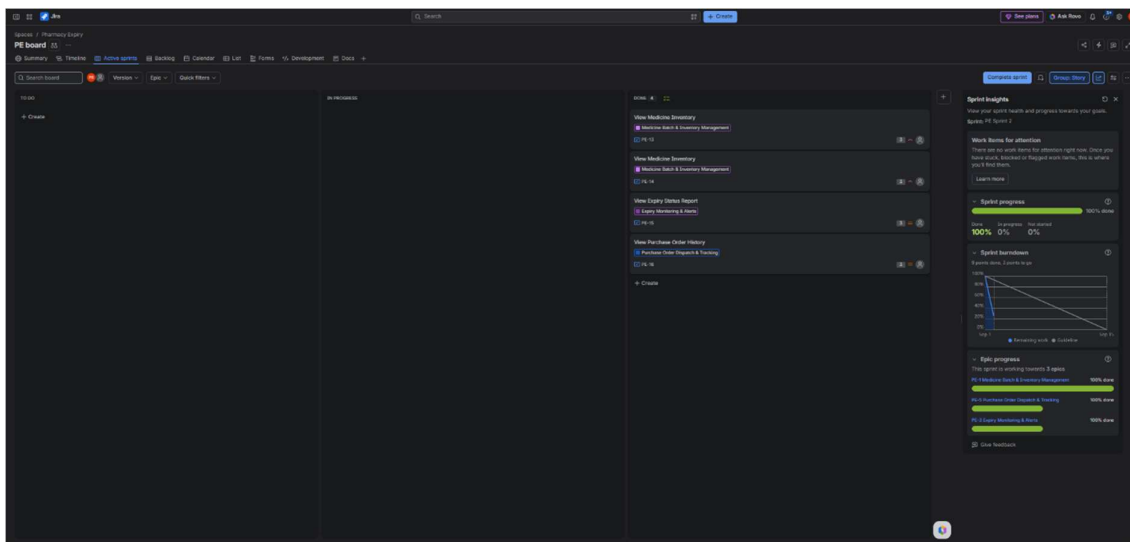
Figure 11: Sprint 2 progress using the Scrum workflow



## 12. Sprint 2 Completion

All Sprint 2 User Stories were moved to Done.

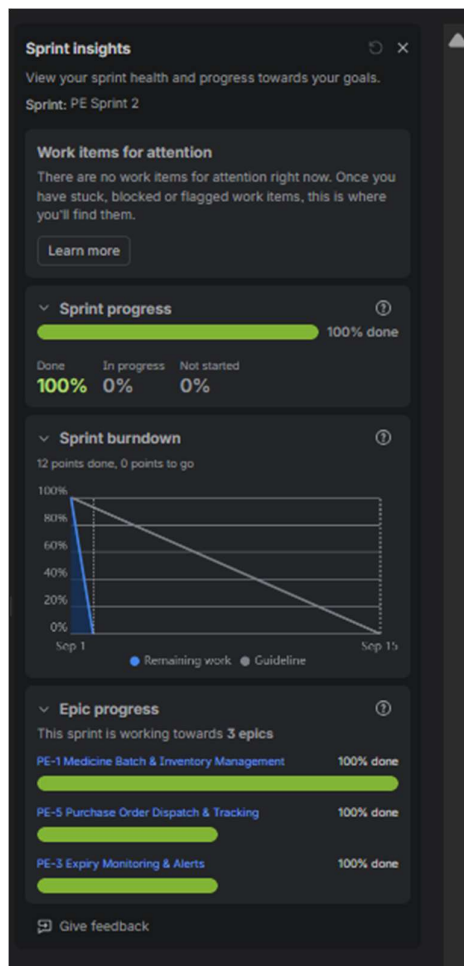
Figure 12: Completion of Sprint 2



## 13. Burndown Chart

Burndown Chart for the 2<sup>nd</sup> sprint

Figure 12: Sprint Burndown Chart showing the progress of completed and remaining Story Points



## 14. Reflection

1. Did your estimations reflect the actual effort?

The Story Points provided a reasonable relative estimate of the effort required for the User Stories. Simulating the Sprints helped compare the estimated effort with the actual progress.

2. Was your backlog well-prioritized?

Yes. The backlog was organized according to user impact and importance, with higher-priority pharmacy operations handled before lower-priority supporting functionality.

3. How did your simulated sprint align with your plan?

The Sprint followed the planned Scrum workflow. User Stories were moved from To Do → In Progress → Done, allowing the team's progress to be tracked throughout the Sprint.

4. What insights did the burndown chart give about your team's capacity?

The Burndown Chart provided a visual representation of the team's remaining workload and progress. It helped determine whether the team was progressing according to the expected Sprint trajectory.